

Semantic Manipulation — Evaluation Tables

xArm · VLA (π -style) · datasets: coffee 1×1 & 4×4, pouring 5×3, mug-tree 5×1 · trained at 50 / ~100 / 150 demos (filled: coffee @50/100/150, pouring 5×3 @50/105/150, pouring 1×1 @50/100/150 + 1×1 SAP baseline, mug-tree @150). Marks: ✓ success, ✗ failure; superscript = grasp (S/T) and note numbers.

Codes. Cups: WB White-Basic, O Orange, Bk Black, R Red (ID); Br Brown, WC White-Ceramic, Gr Gray, P Pink (OOD); TWC Tall-White-Ceramic. Machines: K Keurig, Bk Black, T Teal, W White (ID); R Red, TS Tastyle, Bl Blue (OOD). Bowls: BL Blue, LB Light-Blue, BK Black (ID); P Pink, TB Tall, W White (OOD).

Summary

Table 1: Success rate by condition. Coffee shown at 50/100/150 and pouring at 50/105/150 demos to expose dataset-size scaling; SAP is a secondary baseline on the pouring task.

Run	ID/ID	ID/OOD	OOD/ID	OOD/OOD	Overall
Coffee 4x4 VLA @150	75% _{12/16}	67% _{8/12}	62% _{10/16}	42% _{5/12}	62% _{35/56}
Coffee 4x4 VLA @100	44% _{7/16}	58% _{7/12}	50% _{8/16}	33% _{4/12}	46% _{26/56}
Coffee 4x4 VLA @50	44% _{7/16}	58% _{7/12}	12% _{2/16}	50% _{6/12}	39% _{22/56}
Coffee 1x1 VLA @150	0% _{0/16}	25% _{3/12}	6% _{1/16}	17% _{2/12}	11% _{6/56}
Coffee 1x1 VLA @100	6% _{1/16}	25% _{3/12}	0% _{0/16}	25% _{3/12}	12% _{7/56}
Coffee 1x1 VLA @50	6% _{1/16}	8% _{1/12}	19% _{3/16}	17% _{2/12}	12% _{7/56}
Pouring 5x3 VLA @150	75% _{9/12}	83% _{10/12}	67% _{8/12}	67% _{8/12}	73% _{35/48}
Pouring 5x3 VLA @105	83% _{10/12}	58% _{7/12}	50% _{6/12}	25% _{3/12}	54% _{26/48}
Pouring 5x3 VLA @50	50% _{6/12}	42% _{5/12}	67% _{8/12}	8% _{1/12}	42% _{20/48}
Pouring 1x1 VLA @50 (partial)	0% _{0/8}	50% _{4/8}	25% _{2/8}	25% _{2/8}	25% _{8/32}
Pouring 1x1 VLA @100 (partial)	38% _{3/8}	25% _{2/8}	38% _{3/8}	38% _{3/8}	34% _{11/32}
Pouring 1x1 VLA @150 (partial)	50% _{4/8}	62% _{5/8}	38% _{3/8}	38% _{3/8}	47% _{15/32}
Pouring 1x1 SAP (baseline)	50% _{6/12}	42% _{5/12}	33% _{4/12}	58% _{7/12}	46% _{22/48}

Training runs & dataset sizes

Table 2: Training runs and dataset sizes, cross-listed with *Model Train Baselines*. Each task is trained at 50 / $\sim$ 100 / 150 demos; the eval-done rows match the grids in this report.

Dataset	Size	Train	Started	Eval	xArm
Brick_in_Drawer	50	Done	Last Tuesday	no	✓
Brick_in_Drawer	100	Done	Last Tuesday	no	✓
Brick_in_Drawer	150	Done	June 2, 2026	no	✓
Faucet	50	Done	Last Tuesday	no	✓
Mug Tree Task	50	Done	June 2, 2026	no	✓
Mug Tree Task	100	Done	Last Wednesday	no	✓
Mug Tree Task	150	Done	June 2, 2026	done	✓
Mug/Machine 1by1	50	Done	June 2, 2026	done	✓
Mug/Machine 1by1	100	Done	June 2, 2026	done	✓
Mug/Machine 1by1	150	Done	Last Monday	done	✓
Mug/Machine 4by4	50	Done	Last Monday	done	✓
Mug/Machine 4by4	100	Done	June 2, 2026	done	✓
Mug/Machine 4by4	150	Done	June 2, 2026	done	✓
Pour Task 1by1	50	Done	Yesterday	done	✓
Pour Task 1by1	100	Done	Yesterday	done	✓
Pour Task 1by1	150	Done	Yesterday	done	✓
Pour Task 5by3	50	Done	June 3, 2026	done	✓
Pour Task 5by3	105	Done	June 2, 2026	done	✓
Pour Task 5by3	150	Done	Last Tuesday	done	✓

Table 3: Our-method (SAP) training runs, cross-listed with *Our Method Train*. Only underlined runs carrying a Yes/No in the Eval Completed? column are listed; the done row is the 1 $\times$ 1 pouring model `pour_quality_2k_lerobot` (60%).

Model	Dataset	Size	Status	Started	Eval
Xarm VLA (4by4 mug/cm)	lfg_coffee_machine_scaled_2k	2000	Checkpoint	Today	no
Xarm VLA (r2r2r baseline)	corl_mug_2k_1cm	2000	Checkpoint	Yesterday	no
Xarm VLA	pour_cousins_2k_lerobot	2000	Checkpoint	Yesterday	no
Xarm VLA	pour_quality_2k_lerobot	2000	Done	Tuesday	done

Task 1 — Cup on Coffee Machine (rows: machines, cols: cups)

Table 4: Coffee 4x4 VLA @150 demos — ID Cup / ID Machine (12/16, 75%).

Machine	WB	O	Bk	R
K	✓ ^S	✗ ^S	✓ ^S	✓ ^T
Bk	✓ ^S	✓ ^S	✗ ^T	✓ ^{S 1}
T	✓ ^{T 1}	✓ ^{T 3}	✓ ^{T 1}	✗ ^S
W	✓ ^{T 4}	✓ ^T	✓ ^{T 4}	✗ ^T

Table 5: Coffee 4x4 VLA @150 demos — ID Cup / OOD Machine (8/12, 67%).

Machine	WB	O	Bk	R
R	✓ ^S	✓ ^{T 1}	✓ ^{T 1,5}	✗ ^{S 6}
TS	✓ ^{S 4}	✗ ^S	✗ ^{S 7}	✗ ^S
Bl	✓ ^{S 1}	✓ ^{S 1}	✓ ^T	✓ ^{S 2}

Table 6: Coffee 4x4 VLA @150 demos — OOD Cup / ID Machine (10/16, 62%).

Machine	Br	WC	Gr	P
K	✓ ^{S 2}	✓ ^S	✓ ^S	✓ ^S
Bk	✗ ^{T 8,9}	✓ ^S	✓ ^S	✗ ^S
T	✓ ^{T 1,2,8}	✗ ^{S 10}	✗ ^{T 1,9}	✓ ^S
W	✓ ^{S 2}	✗ ^{T 4}	✓ ^{T 2}	✗ ^{S 6}

Table 7: Coffee 4x4 VLA @150 demos — OOD Cup / OOD Machine (5/12, 42%).

Machine	Br	WC	Gr	P
R	$\times^S$	$\times^S$	$\times^T$	$\checkmark^S$
TS	$\checkmark^{S1}$	$\times^S$	$\checkmark^{S2}$	$\times^S$
Bl	$\times^{S1,7,8}$	$\times^{S8}$	$\checkmark^{S1}$	$\checkmark^S$

Table 8: Coffee 1x1 VLA @150 demos — ID Cup / ID Machine (0/16, 0%).

Machine	WB	O	Bk	R
K	$\times^T$	$\times^{T11}$	$\times^T$	$\times^{T11}$
Bk	$\times^{S8}$	$\times^{T11}$	$\times^T$	$\times^{T11}$
T	$\times^{T1,7}$	$\times^T$	$\times^T$	$\times^{T7}$
W	$\times^{T1,8}$	$\times^{T11}$	$\times^T$	$\times^{S11}$

Table 9: Coffee 1x1 VLA @150 demos — ID Cup / OOD Machine (3/12, 25%).

Machine	WB	O	Bk	R
R	$\checkmark^{T1}$	$\times^S$	$\times^{S11}$	$\times^{S7}$
TS	$\checkmark^S$	$\times^S$	$\times^{T11}$	$\times^{T6}$
Bl	$\checkmark^S$	$\times^{S5}$	$\times^{T11}$	$\times^{T7}$

Table 10: Coffee 1x1 VLA @150 demos — OOD Cup / ID Machine (1/16, 6%).

Machine	Br	WC	Gr	P
K	$\times^{T11}$	$\checkmark^S$	$\times^T$	$\times^{S7,8}$
Bk	$\times^S$	$\times^S$	$\times^T$	$\times^S$
T	$\times^S$	$\times^{T7}$	$\times^{S11}$	$\times^{S7}$
W	$\times^T$	$\times^T$	$\times^T$	$\times^{S10}$

Table 11: Coffee 1x1 VLA @150 demos — OOD Cup / OOD Machine (2/12, 17%).

Machine	Br	WC	Gr	P
R	$\times^{T11}$	$\times^S$	$\times^{S7}$	$\times^{T7,9}$
TS	$\times^S$	$\checkmark^S$	$\times^S$	$\times^T$
Bl	$\checkmark^{S3,5}$	$\times^{T7}$	$\times^{S11}$	$\times^{T5,7}$

Table 12: Coffee 1x1 VLA @150 demos — White-Basic re-run.

Machine	Result	Grasp	Notes
K	✓	S	redo: 1st attempt F
Bk	✗	S	–
T	✓	T	2
W	✗	T	1, 7, 10
R	✓	S	1
TS	✓	S	1
Bl	✗	T	7, 9

Table 13: Coffee 4x4 VLA @100 demos — ID Cup / ID Machine (7/16, 44%).

Machine	WB	O	Bk	R
K	✓ ^S	✗ ^{S 8,10}	✗ ^{S 8}	✓ ^S
Bk	✗ ^{S 4}	✗ ^{S 10}	✗ ^{T 5}	✓ ^S
T	✗ ^{S 11}	✗ ^{S 4,11}	✗ ^{T 11}	✓ ^S
W	✓ ^S	✓ ^S	✓ ^S	✗ ^{T 11}

Table 14: Coffee 4x4 VLA @100 demos — ID Cup / OOD Machine (7/12, 58%).

Machine	WB	O	Bk	R
R	✓ ^S	✓ ^{S 1}	✗ ^{T 11}	✓ ^S
TS	✗ ^{S 11}	✓ ^S	✓ ^S	✗ ^{S 11}
Bl	✓ ^{S 1}	✗ ^{S 8,10}	✗ ^{S 11}	✓ ^S

Table 15: Coffee 4x4 VLA @100 demos — OOD Cup / ID Machine (8/16, 50%).

Machine	Br	WC	Gr	P
K	✗ ^{S 8}	✓ ^S	✗ ^{S 11}	✓ ^{S 2}
Bk	✗ ^{S 4}	✗ ^{S 10,11}	✓ ^{S 1}	✗ ^{S 4,11}
T	✗ ^{S 4,11}	✓ ^T	✗ ^{T 4,11}	✓ ^S
W	✓ ^S	✓ ^S	✓ ^S	✗ ^{S 11}

Table 16: Coffee 4x4 VLA @100 demos — OOD Cup / OOD Machine (4/12, 33%).

Machine	Br	WC	Gr	P
R	$\times$ S 4,11	$\checkmark$ S 1	$\times$ S 11	$\times$ S 11
TS	$\times$ S 2,8	$\checkmark$ S 2	$\times$ S 4,11	$\times$ T 11
Bl	$\times$ S 11	$\times$ 4,11	$\checkmark$ S	$\checkmark$ T

Table 17: Coffee 1x1 VLA @100 demos — ID Cup / ID Machine (1/16, 6%).

Machine	WB	O	Bk	R
K	$\times$ T 5	$\times$ S 10	$\times$ T 11	$\times$ T 11
Bk	$\times$ T 9	$\times$ S 11	$\times$ T 11	$\times$ S 11
T	$\times$ T 1,5	$\times$ S 1,5	$\times$ T 9,11	$\times$ T 7,9,11
W	$\times$ T 7,8	$\checkmark$ S	$\times$ S 11	$\times$ S 4,11

Table 18: Coffee 1x1 VLA @100 demos — ID Cup / OOD Machine (3/12, 25%).

Machine	WB	O	Bk	R
R	$\checkmark$ T 1	$\times$ S 10,11	$\times$ S 11	$\times$ S 11
TS	$\checkmark$ S	$\times$ T 5	$\times$ T 11	$\times$ T 7,11
Bl	$\checkmark$ S	$\times$ S 4,11	$\times$ S 11	$\times$ T 4

Table 19: Coffee 1x1 VLA @100 demos — OOD Cup / ID Machine (0/16, 0%).

Machine	Br	WC	Gr	P
K	$\times$ T 4	$\times$ T 5	$\times$ T 11	$\times$ S 1,5,7
Bk	$\times$ S 11	$\times$ T 9	$\times$ S 9,11	$\times$ T 7,9
T	$\times$ S 11	$\times$ T 4	$\times$ T 11	$\times$ S 1,4,11
W	$\times$ T 11	$\times$ S 12	$\times$ T 9,11	$\times$ S 7,9

Table 20: Coffee 1x1 VLA @100 demos — OOD Cup / OOD Machine (3/12, 25%).

Machine	Br	WC	Gr	P
R	$\times$ S 4	$\checkmark$ T	$\times$ S 11	$\times$ S 7,11
TS	$\times$ T 4	$\times$ S 11	$\times$ S 9,11	$\checkmark$ S
Bl	$\times$ S 9,11	$\checkmark$ S	$\times$ S 11	$\times$ T 5,7,10

Table 21: Coffee 4x4 VLA @50 demos — ID Cup / ID Machine (7/16, 44%).

Machine	WB	O	Bk	R
K	$\times^S 2$	$\times^S 2$	$\times^S$	$\checkmark^S$
Bk	$\times^S 2$	$\times^S 1$	$\checkmark^S$	$\times^S$
T	$\checkmark^S$	$\checkmark^S$	$\times^S 1$	$\times^S 1$
W	$\checkmark^S$	$\checkmark^S$	$\checkmark^T$	$\times^T 2$

Table 22: Coffee 4x4 VLA @50 demos — ID Cup / OOD Machine (7/12, 58%).

Machine	WB	O	Bk	R
R	$\checkmark^S$	$\checkmark^T$	$\times^S 2$	$\checkmark^S$
TS	$\times^S 1$	$\checkmark^S$	$\checkmark^S$	$\checkmark^S$
Bl	$\times^S 2$	$\times^S 1$	$\checkmark^S$	$\times^S 2$

Table 23: Coffee 4x4 VLA @50 demos — OOD Cup / ID Machine (2/16, 12%).

Machine	Br	WC	Gr	P
K	$\times^S$	$\checkmark^S$	$\times^S 5$	$\times^S 6$
Bk	$\times^T 6$	$\times^S 2$	$\times^T 2$	$\times^S 1$
T	$\times^T 2$	$\times^S 2$	$\times^S 2$	$\times^S 2$
W	$\times^S 2$	$\checkmark^S$	$\times^T 2$	$\times^T 1$

Table 24: Coffee 4x4 VLA @50 demos — OOD Cup / OOD Machine (6/12, 50%).

Machine	Br	WC	Gr	P
R	$\checkmark^S 7$	$\checkmark^S 7$	$\times^S 2$	$\checkmark^S$
TS	$\times^S 2$	$\times^S 7,8$	$\times^T 2$	$\times^S 1,8$
Bl	$\checkmark^S$	$\times^S 2,6$	$\checkmark^T 7,9$	$\checkmark^S$

Table 25: Coffee 1x1 VLA @50 demos — ID Cup / ID Machine (1/16, 6%).

Machine	WB	O	Bk	R
K	$\checkmark^T$	$\times^T$	$\times^S$	$\times^T$
Bk	$\times^S 1$	$\times^T 2$	$\times^S 2$	$\times^S 2$
T	$\times^T 2$	$\times^T 2$	$\times^S 2$	$\times^S 1$
W	$\times^T 2$	$\times^S 2$	$\times^T 2$	$\times^T 3$

Table 26: Coffee 1x1 VLA @50 demos — ID Cup / OOD Machine (1/12, 8%).

Machine	WB	O	Bk	R
R	$\times^{T 2}$	$\times^{T 2}$	$\times^{S 2}$	$\times^{T 2}$
TS	$\checkmark^T$	$\times^{S 2}$	$\times^{S 4}$	$\times^{S 2}$
Bl	$\times^{T 5}$	$\times^{S 2}$	$\times^{S 4}$	$\times^{T 2}$

Table 27: Coffee 1x1 VLA @50 demos — OOD Cup / ID Machine (3/16, 19%).

Machine	Br	WC	Gr	P
K	$\times^{T 2}$	$\checkmark^S$	$\times^{T 2}$	$\checkmark^S$
Bk	$\times^{T 2}$	$\times^{T 2}$	$\times^{T 2}$	$\times^{S 2}$
T	$\times^{S 2}$	$\times^{S 2}$	$\times^{S 1,2}$	$\times^{T 10}$
W	$\times^{S 1}$	$\checkmark^{T 9}$	$\times^{T 2}$	$\times^{T 1}$

Table 28: Coffee 1x1 VLA @50 demos — OOD Cup / OOD Machine (2/12, 17%).

Machine	Br	WC	Gr	P
R	$\times^{S 2}$	$\checkmark^T$	$\times^{T 2}$	$\times^{T 2}$
TS	$\times^{S 2}$	$\times^{S 2}$	$\times^{T 2}$	$\times^{S 5}$
Bl	$\times^{S 2}$	$\times^{S 6}$	$\times^{S 2}$	$\checkmark^T$

Task 2 — Cup Pouring (rows: bowls, cols: cups)

Table 29: Pouring 5x3 VLA @150 demos — ID Cup / ID Bowl (9/12, 75%).

Bowl	WB	O	Bk	R	TWC
BL	✓ ^S	✓ ^{T 1}	✓ ^S	✓ ^S	—
LB	✓ ^T	✗ ¹⁶	✓ ^T	✓ ^{S 20}	—
BK	✓ ^{S 21}	✓ ^T	✗ ^{16,18}	✗ ¹⁶	—

Table 30: Pouring 5x3 VLA @150 demos — ID Cup / OOD Bowl (10/12, 83%).

Bowl	WB	O	Bk	R	TWC
P	✗ ¹⁶	✓ ^T	✓ ^T	✓ ^T	—
TB	✓ ^{S 6}	✓ ^{S 6}	✗ ^{T 6,20}	✓ ^{T 6}	—
W	✓ ^S	✓ ^T	✓ ^T	✓ ^{S 3}	—

Table 31: Pouring 5x3 VLA @150 demos — OOD Cup / ID Bowl (8/12, 67%).

Bowl	Br	WC	Gr	P
BL	✓ ^{S 3}	✓ ^{T 20}	✗ ¹⁶	✗ ¹⁶
LB	✓ ^T	✓ ^{S 18}	✓ ^{T 13,15}	✗ ^{S 20}
BK	✓ ^S	✗ ¹⁶	✓ ^T	✓ ^{T 3}

Table 32: Pouring 5x3 VLA @150 demos — OOD Cup / OOD Bowl (8/12, 67%).

Bowl	Br	WC	Gr	P
P	✓ ^T	✗ ^{S 20}	✓ ^T	✗ ^{T 14,21}
TB	✓ ^{T 6}	✗ ^{T 20}	✗ ^{T 20}	✓ ^{T 3}
W	✓ ^T	✓ ^S	✓ ^T	✓ ^T

Table 33: Pouring 5x3 VLA @105 demos — ID Cup / ID Bowl (10/12, 83%).

Bowl	WB	O	Bk	R	TWC
BL	✓ ^S	✓ ^S	✓ ^{T 1}	✓ ^{S 2}	✓ ^{T 3}
LB	✓ ^{S 2,3}	✓ ^T	✗ ^T	✓	✓ ^{STS}
BK	✓ ^S	✓ ^T	✓	✗	✓ ^{T 4}

Table 34: Pouring 5x3 VLA @105 demos — ID Cup / OOD Bowl (7/12, 58%).

Bowl	WB	O	Bk	R	TWC
P	✓ ^{S 3}	✗	✓ ^S	✓ ^S	✓ ^{S 6}
TB	✗	✗ ⁷	✗ ¹¹	✓ ^{T 8}	✗
W	✗ ¹¹	✓ ^{S 9}	✓ ^S	✓ ^{T 8}	✓ ^{T 2}

Table 35: Pouring 5x3 VLA @105 demos — OOD Cup / ID Bowl (6/12, 50%).

Bowl	Br	WC	Gr	P
BL	✗	✗ ¹⁰	✗ ¹¹	✓ ^S
LB	✓ ^S	✓ ^{S 2}	✓ ^S	✗ ¹¹
BK	✗ ^{5,6}	✓ ^{T 3}	✓ ^S	✗

Table 36: Pouring 5x3 VLA @105 demos — OOD Cup / OOD Bowl (3/12, 25%).

Bowl	Br	WC	Gr	P
P	✗ ¹¹	✗ ¹⁰	✓	✓ ^{T 2}
TB	✗ ¹¹	✗ ¹¹	✗ ¹¹	✗ ⁸
W	✓	✗ ¹¹	✗ ¹²	✗ ¹²

Table 37: Pouring 5x3 VLA @50 demos — ID Cup / ID Bowl (6/12, 50%).

Bowl	WB	O	Bk	R	TWC
BL	$\times$ S 13,14,15	$\checkmark$ T	$\checkmark$ S 3	$\times$ ¹⁶	—
LB	$\times$ ¹⁶	$\times$ T 1,3,17,18	$\times$ T 1,3,17,18	$\checkmark$ S	—
BK	$\checkmark$ T 11	$\times$ ⁵	$\checkmark$ T 2	$\checkmark$ S	—

Table 38: Pouring 5x3 VLA @50 demos — ID Cup / OOD Bowl (5/12, 42%).

Bowl	WB	O	Bk	R	TWC
P	$\checkmark$ S 1,14,17	$\times$ S 19	$\times$ T 20	$\times$ T 20	—
TB	$\times$ T 20	$\checkmark$ S 6,14	$\times$ S 20	$\checkmark$ T 6	—
W	$\checkmark$ S 11	$\checkmark$ T	$\times$ S 18	$\times$ ⁷	—

Table 39: Pouring 5x3 VLA @50 demos — OOD Cup / ID Bowl (8/12, 67%).

Bowl	Br	WC	Gr	P
BL	$\checkmark$ T 3	$\times$ T 20	$\times$ ^{7,21}	$\checkmark$ S 1
LB	$\times$ ⁵	$\times$ ⁷	$\checkmark$ T 6	$\checkmark$ S
BK	$\checkmark$ T 3	$\checkmark$ S 1,3	$\checkmark$ T 12	$\checkmark$ S

Table 40: Pouring 5x3 VLA @50 demos — OOD Cup / OOD Bowl (1/12, 8%).

Bowl	Br	WC	Gr	P
P	$\times$ T 20	$\times$ T 18	$\times$ ^{14,18}	$\times$ ^{5,16}
TB	$\times$ ^{5,16,18}	$\times$ ¹⁸	$\times$ ²⁰	$\times$ ¹⁶
W	$\times$ ^{16,18,21}	$\times$ T 11,18	$\times$ ²⁰	$\checkmark$ T 11,21

Table 41: Pouring 1x1 VLA @50 demos (partial eval) — ID Cup / ID Bowl (0/8, 0%).

Bowl	WB	O	Bk	R	TWC
BL	$\times$ ^{13,18}	$\times$ S 6,20	$\times$ ^{5,16}	$\times$ ^{5,16}	—
LB	$\times$ ¹⁶	—	$\times$ ^{5,16}	—	—
BK	—	$\times$ ^{5,16}	—	$\times$ ^{5,16}	—

Table 42: Pouring 1x1 VLA @50 demos (partial eval) — ID Cup / OOD Bowl (4/8, 50%).

Bowl	WB	O	Bk	R	TWC
P	✓ ^{T 2}	—	✗ ¹⁶	✓ ^{T 6}	—
TB	—	✓ ^{T 6}	✗ ^{5,16}	✗ ^{14,16}	—
W	✓ ^T	✗ ^{T 11}	—	—	—

Table 43: Pouring 1x1 VLA @50 demos (partial eval) — OOD Cup / ID Bowl (2/8, 25%).

Bowl	Br	WC	Gr	P
BL	✗ ^{5,16}	✗ ^{S 11}	✗ ^{5,16}	✓ ^{T 3}
LB	✗ ^{5,16}	✓ ^{T 6}	—	—
BK	—	—	✗ ^{5,16}	✗ ^{5,16}

Table 44: Pouring 1x1 VLA @50 demos (partial eval) — OOD Cup / OOD Bowl (2/8, 25%).

Bowl	Br	WC	Gr	P
P	✗ ^{5,16}	—	✗ ^{5,16}	✓ ^T
TB	—	✗ ^{T 6,20}	✗ ^{5,16}	✗ ^{5,16}
W	✓ ^{T 3}	✗ ^{5,16}	—	—

Table 45: Pouring 1x1 VLA @100 demos (partial eval) — ID Cup / ID Bowl (3/8, 38%).

Bowl	WB	O	Bk	R	TWC
BL	✓ ^S	✓ ^T	—	—	—
LB	—	—	✗ ^{5,16}	✗ ^{T 6,20}	—
BK	✗ ^{13,16,18}	✓ ^T	✗ ^{16,18}	✗ ^{16,18}	—

Table 46: Pouring 1x1 VLA @100 demos (partial eval) — ID Cup / OOD Bowl (2/8, 25%).

Bowl	WB	O	Bk	R	TWC
P	✗ ^{13,16,18}	—	✗ ¹⁶	✗ ^{S 19,20}	—
TB	—	✗ ^{13,16,18}	✗ ^{5,16}	—	—
W	✗ ^{T 11}	✓ ^{T 3}	—	✓ ^{T 1,6}	—

Table 47: Pouring 1x1 VLA @100 demos (partial eval) — OOD Cup / ID Bowl (3/8, 38%).

Bowl	Br	WC	Gr	P
BL	✓ ^{T 3}	✗ ²⁰	✗ ^{18,20}	—
LB	✗ ^{T 20}	—	—	✓ ^{T 6,20}
BK	—	✓ ^{S 3}	✗ ¹⁶	✗ ^{S 6,11,20}

Table 48: Pouring 1x1 VLA @100 demos (partial eval) — OOD Cup / OOD Bowl (3/8, 38%).

Bowl	Br	WC	Gr	P
P	✓ ^T	—	✗ ¹⁶	✓ ^S
TB	—	✗ ^{5,16}	✗ ^{5,16}	✓ ^{T 3,6}
W	✗ ^{14,21}	✗ ^{T 6,11}	—	—

Table 49: Pouring 1x1 VLA @150 demos (partial eval) — ID Cup / ID Bowl (4/8, 50%).

Bowl	WB	O	Bk	R	TWC
BL	✗ ^{T 20}	✗ ^{S 20}	✓ ^{S 6,20}	✗ ^{T 11}	—
LB	✓ ^S	✓ ^{T 1}	—	—	—
BK	—	—	✗ ^{S 5}	✓ ^{T 3}	—

Table 50: Pouring 1x1 VLA @150 demos (partial eval) — ID Cup / OOD Bowl (5/8, 62%).

Bowl	WB	O	Bk	R	TWC
P	✓ ^T	—	✗ ^{16,18}	—	—
TB	✓ ^{T 6}	✗ ^{T 16}	✗ ^{5,16}	✓ ^S	—
W	—	✓ ^{S 3}	—	✓ ^S	—

Table 51: Pouring 1x1 VLA @150 demos (partial eval) — OOD Cup / ID Bowl (3/8, 38%).

Bowl	Br	WC	Gr	P
BL	✓ ^T	—	✓ ^{T 3}	✗ ^{T 11}
LB	✗ ^{T 6,20}	✓ ^T	—	✗ ^{14,16}
BK	—	✗ ¹⁶	✗ ^{T 6,11}	—

Table 52: Pouring 1x1 VLA @150 demos (partial eval) — OOD Cup / OOD Bowl (3/8, 38%).

Bowl	Br	WC	Gr	P
P	$\times^{T16}$	—	—	$\checkmark^S$
TB	—	$\times^{S11}$	$\times^{16,21}$	$\times^{T11}$
W	$\checkmark^S$	$\checkmark^{T3}$	$\times^{S11}$	—

Task 3 — Hang Cup on Mug Tree (5 trials per cup)

Table 53: Mug Tree 5x1 VLA @150 demos — ID cups.

Cup	1	2	3	4	5	Rate
WB	✓	✓	✓	✓	✓	100% _{5/5}
O	✓ ¹	✗ ^{1,2}	✓ ¹	✗ ^{1,2}	✓	60% _{3/5}
Bk	✗ ³	✗ ^{3,4}	✗ ^{3,4}	✗ ³	✗ ³	0% _{0/5}
R	✓	✓	✓ ⁴	✗ ³	✓	80% _{4/5}
TWC*	✗ ²	✓	✓ ¹	✗ ²	✗ ³	40% _{2/5}

Table 54: Mug Tree 5x1 VLA @150 demos — OOD cups.

Cup	1	2	3	4	5	Rate
Br	✓	✗ ^{1,3}	✓	✓ ¹	✓ ⁵	80% _{4/5}
WC	✗ ³	✗ ⁶	✓	✗ ⁷	✓	40% _{2/5}
Gr	✓ ¹	✓ ¹	✓ ¹	✗ ²	✗	60% _{3/5}
P	✗ ²	✓	✓	✗	✗	40% _{2/5}
Pu	✓	✓	✓	✗	✗ ²	60% _{3/5}

Task 2b — Cup Pouring, 1x1 SAP baseline (rows: bowls, cols: cups)

Table 55: Pouring 1x1 SAP baseline — ID Cup / ID Bowl (6/12, 50%).

Bowl	WB	O	Bk	R	TWC
BL	✓ ^S	✗ ^{1,2}	✓ ^S	✗ ³	—
LB	✗ ^{S 3,4}	✓ ^S	✗ ³	✓ ^{S 6}	—
BK	✓ ^{S 6}	✗ ³	✗ ³	✓ ^{S 4}	—

Table 56: Pouring 1x1 SAP baseline — ID Cup / OOD Bowl (5/12, 42%).

Bowl	WB	O	Bk	R	TWC
P	✓ ^S	✗ ³	✓ ^S	✓ ^{S 7}	—
TB	✗ ^{S 8}	✗ ³	✗ ³	✗ ^{S 1,8}	—
W	✓ ^S	✗ ³	✗ ³	✓ ^{S 9}	—

Table 57: Pouring 1x1 SAP baseline — OOD Cup / ID Bowl (4/12, 33%).

Bowl	Br	WC	Gr	P
BL	✗ ³	✓ ^S	✗ ³	✗ ³
LB	✓ ^S	✓ ^S	✗ ³	✗ ³
BK	✗ ³	✓ ^S	✗ ³	✗ ³

Table 58: Pouring 1x1 SAP baseline — OOD Cup / OOD Bowl (7/12, 58%).

Bowl	Br	WC	Gr	P
P	✗ ³	✓ ^S	✓ ^{S 6}	✗ ³
TB	✗ ^{S 1,8}	✓ ^{S 11}	✓ ^{S 6}	✗ ³
W	✗ ³	✓ ^{S 6}	✓ ^{S 4}	✓ ^S

Appendix A — Coffee 4x4 VLA @150 demos (OLD run)

Table 59: Coffee 4x4 VLA OLD @150 demos — ID Cup / ID Machine (15/16, 94%).

Machine	WB	O	Bk	R
K	✓ ^T	✓ ^S	✗ ^T	✓ ^T
Bk	✓ ^S	✓ ^T	✓ ^S	✓ ^T
T	✓ ^S	✓ ^S	✓ ^S	✓ ^T
W	✓ ^S	✓ ^T	✓ ^T	✓ ^T

Table 60: Coffee 4x4 VLA OLD @150 demos — ID Cup / OOD Machine (7/8, 88%).

Machine	WB	O	Bk	R
R	✓ ^S	✓ ^T	✓ ^T	✓ ^T
TS	✗ ^T	✓ ^S	✓ ^T	✓ ^T
Bl	—	—	—	—

Table 61: Coffee 4x4 VLA OLD @150 demos — OOD Cup / ID Machine (16/16, 100%).

Machine	Br	WC	Gr	P
K	✓ ^S	✓ ^{T O}	✓ ^T	✓ ^S
Bk	✓ ^T	✓ ^S	✓ ^T	✓ ^S
T	✓ ^S	✓ ^S	✓ ^T	✓ ^S
W	✓ ^T	✓ ^T	✓ ^T	✓ ^S

Table 62: Coffee 4x4 VLA OLD @150 demos — OOD Cup / OOD Machine (8/8, 100%).

Machine	Br	WC	Gr	P
R	✓T	✓S*	✓T	✓S
TS	✓T	✓S**	✓T	✓S
Bl	—	—	—	—

Footnote legends

Coffee notes (150-demo & OLD sheets). **1** Pushed machine back; **2** Tilted placement and contact on plate; **3** Took double time; **4** Pushed cup while trying to grasp; **5** Dropped cup on edge of platform; **6** Self collision; **7** Spilled cup; **8** Force-torque trigger; **9** Grasped inside cup; **10** Gripper hooked on handle; **11** Cannot lift mug; **O** Completed but took ≥ 1 min; * Not fully in cup holder; ** Released while tilted

Coffee notes (100-demo sheets). **1** Pushed machine back; **2** Tilted placement and contact on plate; **3** Took double time; **4** Pushed cup while trying to grasp; **5** Dropped cup on edge of platform; **6** Self collision; **7** Spilled cup; **8** Force-torque trigger; **9** Grasped inside cup; **10** Gripper hooked on handle; **11** Cannot grab handle; **12** Tried to grab entire cup

Coffee notes (50-demo sheets). **1** Spilled the cup; **2** Failed to grasp (hit cup, grasped air); **3** Hit the coffee machine; **4** Good pose, high Z; **5** Wrong grasp; **6** Torque sensor error; **7** Moved machine back; **8** Handle hooked on gripper; **9** Balancing on edge; **10** Dropped cup elsewhere

Pouring notes (VLA: 50/105/150-demo sheets). **1** Bad return cup to table; **2** Went in for second pour; **3** A bit off-center for pour; **4** Tipped bowl while returning cup; **5** Started pushing cup down & didn't move; likely F/T error; **6** Tapped bowl while placing down or pouring; **7** Picked up cup and stopped moving; **8** Tall bowl was pushed while attempting to pour; **9** Two partial pours - ≥ 1 completed; **10** Gripper opened while pour attempt; **11** Severely off center; **12** Tipped cup over while picking up cup; **13** Gripper hooked on handle; **14** Spilled out of bowl / spilled cup; **15** Can't place cup after done; **16** Didn't pick up cup; **17** Poured but didn't return cup properly; **18** F/T error; **19** Dripped cup in bowl; **20** Barely poured; **21** Picked up inside of cup; **STS** Succeeded only after trying all three grasp attempts (side - $\geq$ top - $\geq$ side)

Pouring notes (SAP baseline sheet). **1** Spilled cup; **2** F/T error; **3** Missed handle; **4** Self collision; **5** Tapped the bowl during pour; **6** Off-center pour; **7** Bad return; **8** Grabbed inside of cup; **9** Slightly moved cup; **10** Hooked on handle while grasping; **11** Barely poured

Mug-tree notes. **1** Pushed the tree; **2** Hung on edge, so dropped cup; **3** Dropped mug while not on branch; **4** Tilted grasp on mug lip; **5** Hanging mug on edge; **6** Arm errored out; **7** Tried to grab entire cup